

LegalGPT-India++

A Curriculum Learning and QLoRA-Based Multi-Domain Legal Question Answering Framework for Indian Constitutional, Criminal, and Procedural Law

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Presentation Outline

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Introduction

- Large Language Models (LLMs) have transformed Natural Language Processing.
- Legal documents contain complex domain-specific terminology.
- General-purpose LLMs often struggle with legal reasoning.
- Indian law consists of constitutional, criminal, and procedural provisions.
- Legal document interpretation requires expertise.
- Existing legal AI systems focus largely on Western jurisdictions.
- Explainability is essential in legal applications.
- LegalGPT-India++ aims to address these challenges.

Goal: Build an explainable legal AI assistant specialized for Indian law.

Problem Statement and Motivation

Major Challenges in Legal AI

- Lack of Indian-law-specific language models.
- High computational cost of LLM fine-tuning.
- Need for trustworthy and explainable outputs.
- Difficulty understanding legal provisions.
- Requirement for legal-domain awareness.
- Limited availability of Indian legal QA datasets.
- Manual legal search consumes significant time.
- Citizens often lack access to legal assistance.

Motivation:

Develop a scalable, explainable, and efficient Indian legal question answering system.

Primary Objectives

- 1 Develop LegalGPT-India++.
- 2 Fine-tune Llama-3-8B using QLoRA.
- 3 Utilize Unsloth optimization framework.
- 4 Build a multi-domain legal corpus.
- 5 Implement curriculum learning.
- 6 Introduce weighted dataset generation.
- 7 Enable explainable legal responses.
- 8 Evaluate semantic and legal performance.

Major Contributions

- 1 Construction of a legal corpus containing 12,296 QA pairs.
- 2 Integration of Constitution, IPC, and CrPC datasets.
- 3 Adoption of curriculum learning.
- 4 Implementation of weighted legal dataset engineering.
- 5 Efficient QLoRA-based Llama-3 adaptation.
- 6 Explainable legal source attribution.
- 7 Domain-aware legal response generation.
- 8 Achievement of 92% domain classification accuracy.

Outcome:

A specialized legal language model for Indian law.

Related Work Overview

Approach		Model	Major Limitation
Legal QA Systems		Rule-Based	Limited contextual understanding
Legal-BERT	Legal	Transformer	Western legal focus
GPT-Based Assistants		LLMs	Limited explainability
RAG-Based QA	Legal	Retriever + LLM	Complex architecture
Domain LLMs		Various	Limited Indian legal support

Research Gap

No existing framework simultaneously combines:

- Curriculum Learning
- QLoRA
- Explainable Outputs

Training Corpus Composition

- Multi-domain Indian legal dataset.
- Total of 12,296 legal Question-Answer pairs.
- Covers Constitutional Law, IPC, and CrPC.
- Designed for Legal Question Answering.
- Includes domain labels.
- Includes source attribution labels.
- Supports explainable legal AI.
- Supports curriculum-based learning.

Purpose:

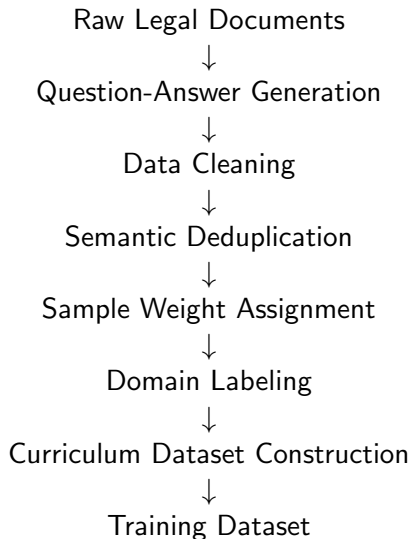
To create a domain-specific legal corpus for training LegalGPT-India++.

Dataset Statistics

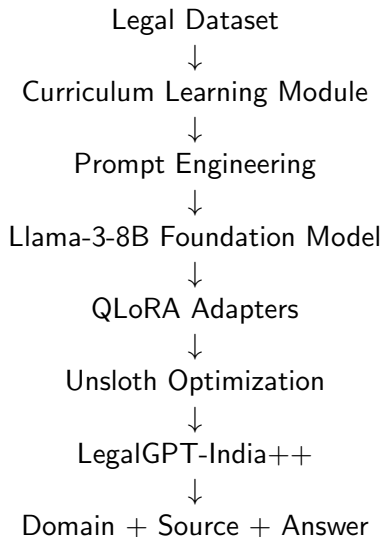
Dataset	Samples	Percentage
Constitution	3435	27.94%
IPC	2015	16.39%
CrPC	6846	55.67%
Total	12296	100%

- CrPC represents the largest legal domain.
- Constitution provides foundational legal knowledge.
- IPC contributes criminal law concepts.
- Multi-domain coverage improves generalization.

Data Preprocessing Pipeline



LegalGPT-India++ Architecture



Foundation Model and Fine-Tuning Strategy

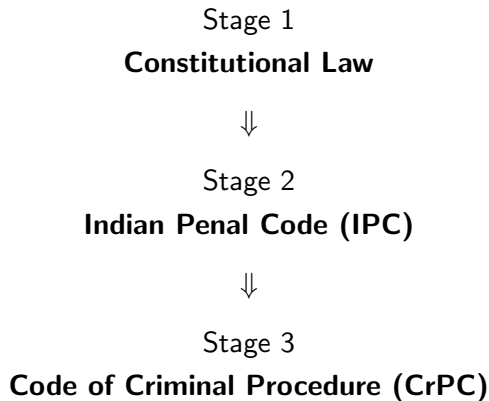
Llama-3-8B

- Foundation Large Language Model.
- Strong language understanding.
- Scalable architecture.
- Suitable for domain adaptation.

QLoRA + Unsloth

- Reduces GPU memory usage.
- Enables efficient fine-tuning.
- Faster training.
- Lower computational cost.

Curriculum Learning Framework



Benefits:

- Progressive knowledge acquisition.
- Stable learning process.

Training Prompt Format

Question:

What is Article 14?

Predicted Domain:

Constitutional Law

Legal Source:

Constitution

Answer:

Article 14 guarantees equality before law and equal protection of laws.

- Standardized prompt structure.
- Encourages explainability.
- Improves domain learning.
- Enables source attribution.

Experimental Setup

Parameter	Value
Base Model	Llama-3-8B
Fine-Tuning Method	QLoRA
Optimization Framework	Unsloth
Dataset Size	12,296 Samples
Training Epochs	1
Domains	Constitution, IPC, CrPC

- Parameter-efficient training strategy.
- Multi-domain legal dataset.
- Curriculum learning enabled.
- Weighted dataset utilized.
- Explainable prompting incorporated.

Evaluation Metrics

Metrics Used

- BLEU Score
- ROUGE-1
- ROUGE-2
- ROUGE-L
- BERTScore Precision
- BERTScore Recall
- BERTScore F1
- Domain Classification Accuracy

Accuracy Formula

$$\text{Accuracy} = \frac{\text{Correct Predictions}}{\text{Total Predictions}} \times 100$$

Quantitative Results

Metric	Score
BLEU	0.208
ROUGE-1	0.531
ROUGE-2	0.428
ROUGE-L	0.481
BERT Precision	0.869
BERT Recall	0.923
BERT F1	0.895
Domain Accuracy	92%

Observation:

Strong semantic performance with high domain-awareness.

Training Analysis and Explainability

- Initial training loss approximately 1.60.
- Final training loss reduced to 1.456.
- Stable convergence observed.
- No major optimization instability.
- Explainable prompting enabled.
- Domain information generated.
- Legal source attribution supported.
- Improved transparency of legal responses.

$$Loss_{Reduction} = 1.60 - 1.456 = 0.144$$

Advantages, Limitations and Future Work

Advantages

- Explainable outputs.
- Computational efficiency.
- Legal domain awareness.
- Scalable architecture.

Limitations

- Dataset imbalance.
- Single epoch training.
- Hallucination possibility.
- Limited legal coverage.

Future Directions

- Retrieval-Augmented Generation (RAG).
- Judicial precedent integration.
- Multilingual legal support.
- Advanced legal reasoning.

Conclusion

- Developed LegalGPT-India++ for Indian legal question answering.
- Constructed a multi-domain legal corpus of 12,296 QA pairs.
- Applied curriculum learning for progressive legal knowledge acquisition.
- Fine-tuned Llama-3-8B using QLoRA and Unsloth.
- Introduced explainable legal response generation.
- Achieved strong semantic performance.
- Obtained BERTScore F1 of 0.895.
- Achieved domain classification accuracy of 92%.
- Established a scalable framework for future legal AI research.

Thank You

Questions and Discussion